Introduction

CNS | ALA-3

NAME

Nandani Makwana

ENROLLMENT NO.

241105060077

ASSIGNMENT

CNS ALA-3



Social Engineering Attacks & Major Security Breaches

The biggest breaches rarely start with code. They start with a convincing lie.

THE ATTACK LIFECYCLE



Research

Scout the target's name, role, habits

Contact

Reach out by email, phone, or in person

Build Trust

Impersonate someone familiar or in charge

Exploit

Get a click, a password, or a plugged-in drive

Cover Tracks

Delete logs, keep access quiet

Three Ways In

Different lies, same goal: get a human to open the door.



Phishing

A fake message impersonates a trusted sender to harvest credentials or spread malware.

Email lure → Fake login page → Credentials stolen



Pretexting

An attacker invents a believable scenario — “IT support,” “a vendor” — to extract information.

Fabricated role → Earned trust → Data handed over



Baiting

A tempting physical or digital item — a USB drive, a free download — hides malware inside.

Tempting item → Curiosity → Malware runs

When It Worked: Real Breaches

Three documented incidents, each built on a different technique.

PHISHING

40M

card numbers stolen

Target, 2013

Attackers phished Target's HVAC vendor for network credentials, then installed malware on checkout systems nationwide.

Could've been stopped: Isolate vendor access from payment networks.

PRETEXTING

130

accounts hijacked

Twitter, 2020

Callers posing as IT support tricked employees into a fake VPN login page, harvesting credentials used to seize verified accounts.

Could've been stopped: Verify IT requests through a separate, known channel.

BAITING

14

months to eradicate

Buckshot Yankee, 2008

A malware-laden USB drive left in a parking lot was plugged into a military laptop, letting a worm spread across classified networks.

Could've been stopped: Ban unknown removable media; disable autorun.

Building Human Firewalls

Technology stops some attacks. Habits stop the rest.

Verify independently

Call back on a number you already know — never one given by the requester.

Pause before clicking

Check the sender address and link destination before acting.

Limit public details

Less personal info online means fewer believable pretexts.

Segment the network

Keep vendor and guest access away from critical systems.

Train and simulate

Regular phishing drills build instinct, not just awareness.

Report immediately

Fast reporting shrinks the time attackers have to act.